

fork_wall / no-aux agent: extended experiments

training dynamics, skill use, generalisation, belief geometry,
probe-calibrated steering, and advance planning

All experiments use the reward-only (no auxiliary loss) PPO+GRU agent from the `fork_wall` chapter (GRU 128; success 0.99). Evaluation maps are held-out seeds; policies are sampled stochastically throughout.

The agent architecture

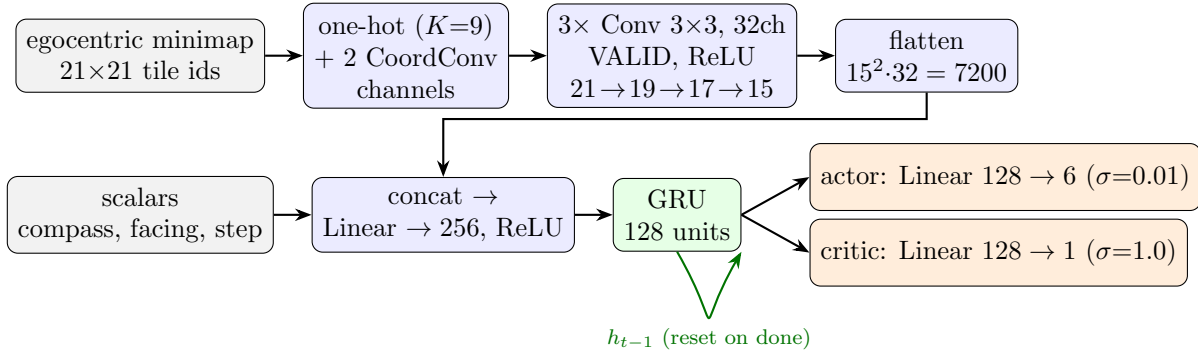


Figure 1: The PPO+GRU policy (PPOGRUPolicy, `obs_encoding=onehot`). The 21×21 egocentric tile crop is one-hot encoded, two CoordConv channels (normalised row/column) are appended, and three unpadded 3×3 convolutions extract spatial features that are flattened, concatenated with the scalar features, and compressed to a 256-d embedding. A single 128-unit GRU is the agent’s only memory: every analysis in this report (probes, belief manifold, clamps) operates on its output h_t . Actor and critic are *linear* read-outs of h_t — so the linear-probe family used throughout is exactly the function class the policy itself uses to read its own state. Actions: 4 movements + **build** + **mine**. The same architecture (ported to JAX, view 7×7, 3 actions) is used for the KeyDoor agents.

1 Training curves (no-aux)

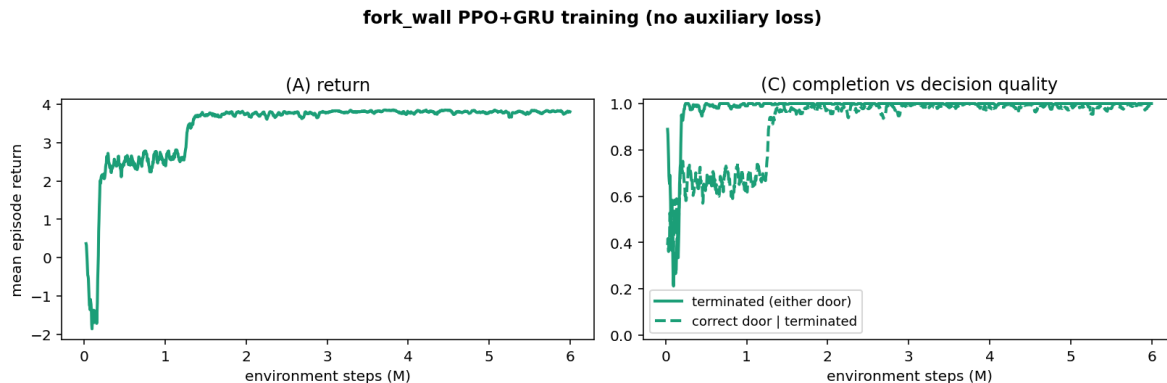


Figure 2: (A) return and (C) completion/decision split. Task completion (solid: terminated at either door) saturates at 1.0 within the first ~ 1 M steps; decision quality (dashed: correct door among terminated) is the slow component that drives the remaining learning.

2 Skill use by category

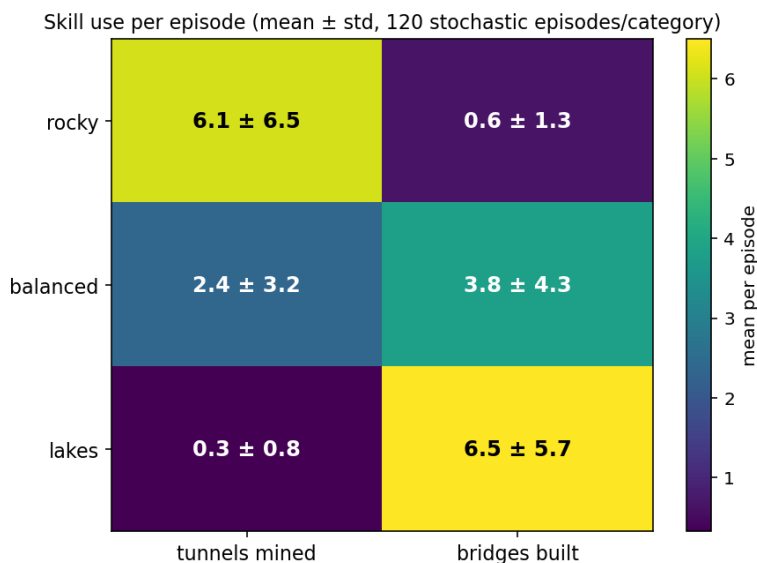


Figure 3: Tunnels mined / bridges built per episode (mean $\pm$ std, 120 stochastic episodes per category, held-out maps). Behaviour is category-appropriate — mining dominates on **rocky**, bridging on **lakes**, a mix on **balanced** — with large episode-to-episode spread reflecting the map-dependent (and often skill-free, detouring) crossing style.

3 Out-of-distribution generalisation

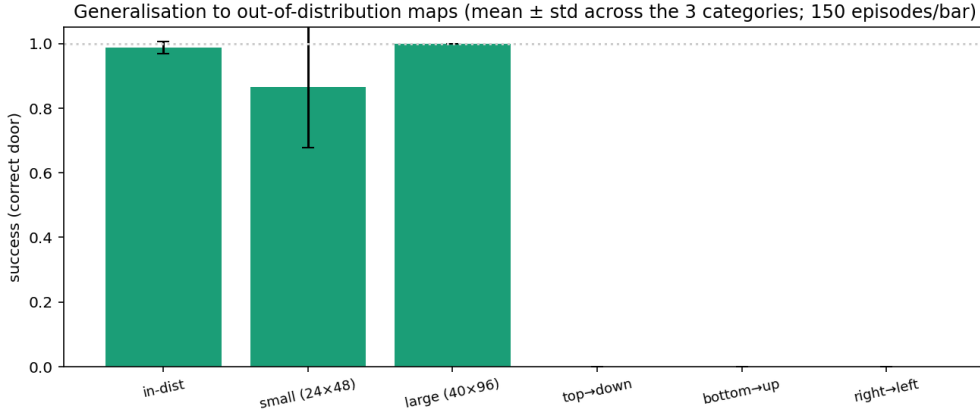


Figure 4: Success on OOD map families, averaged across the three categories (error bars: std across categories; 150 episodes/bar). The agent transfers perfectly to larger maps and nearly so to smaller ones, but fails *completely* on all three rotations.

The rotation failure is directional overfitting, not a broken evaluation. The rotated environments are valid (the potential-based shaping, goal handling and observations are all map-record-driven, and the goal compass rotates consistently). Trajectory inspection shows what happens: on a top→down map the agent still marches *east* until it pins against the right wall and idles there to timeout (0/64 rows travelled toward the goal; 0% of episodes reach any door). Trained exclusively on west→east maps, the policy has compiled “progress = go east” into its behaviour and ignores the compass observation that points elsewhere — a clean example of a habitized spatial prior overriding an informative input.

4 Belief formation

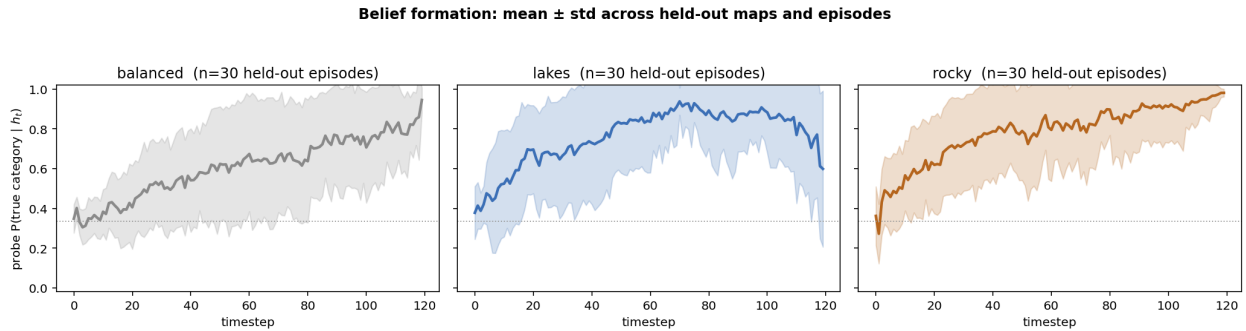


Figure 5: Probe belief $P(\text{true category} \mid h_t)$, mean $\pm$ std over all held-out episodes of each category (linear probe trained on disjoint episodes). The belief is a genuine *accumulation*: it starts at chance ($1/3$) and climbs over ~ 100 steps as terrain evidence enters the view, reaching ~ 0.9 late in the episode. **rocky** rises fastest, **balanced** — the distribution that overlaps both alternatives — most slowly and with the widest variance; the late **lakes** dip reflects the few episodes that survive long past the typical decision time.

5 End-of-episode belief and door outcome

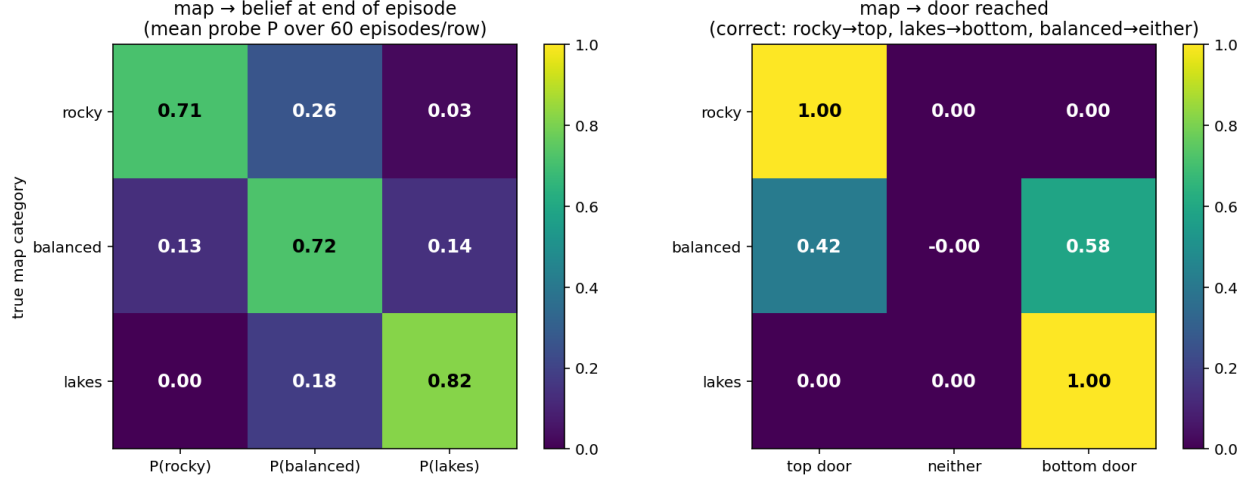


Figure 6: Map → belief and map → door matrices (held-out maps, 60 stochastic episodes per row). *Left:* mean probe belief at the final timestep — diagonal-dominant (0.71–0.82), with the residual mass concentrated on **balanced**, the distribution that genuinely overlaps both alternatives; **rocky** and **lakes** are never confused with each other (≤ 0.03). *Right:* door reached — behaviour is exact: **rocky**→top and **lakes**→bottom at 1.00, and on **balanced** maps (either door correct) the agent splits 0.42/0.58. The belief matrix is graded while the door matrix is crisp: the policy thresholds its graded belief into a discrete choice.

6 Belief manifold

Belief manifold: 3D PCA of GRU states (EV 20%/13%/10%) with the mean hidden-state trajectory $t=0 \rightarrow 80$ (●=start, ★=t80)

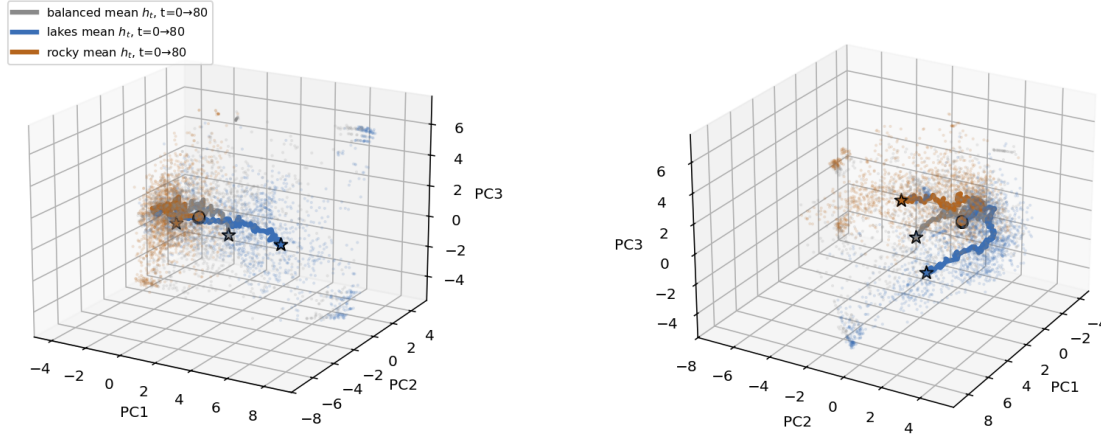


Figure 7: 3D PCA of GRU states (two viewpoints), scatter = individual timesteps coloured by category; bold curves = the *mean hidden state* at each timestep $t = 0 \dots 80$ per category (● = start, ★ = $t=80$). All categories launch from a common origin and separate onto category-specific branches as terrain evidence accumulates — the population-level picture of the belief formation curves of the previous figure.

7 Linear probes with uncertainty

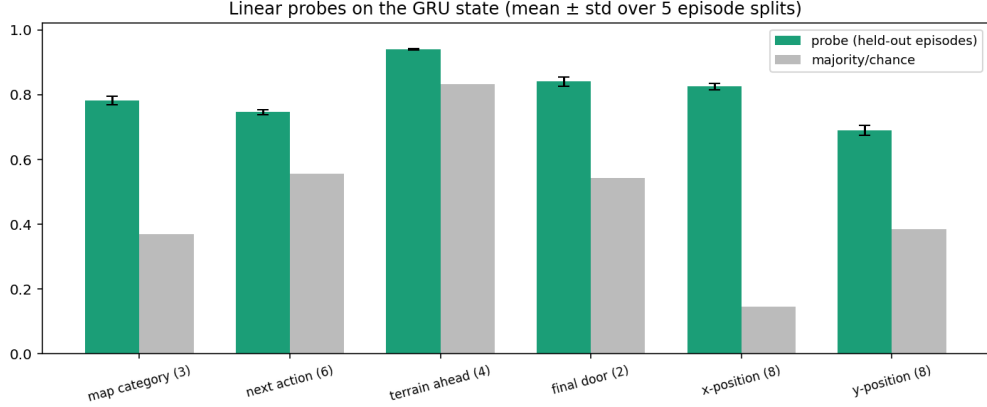


Figure 8: Linear probe battery on the GRU state (mean $\pm$ std over 5 random episode-level train/test splits; grey = majority/chance). The reward-only agent linearly encodes the map category, both position coordinates, the local terrain ahead, its next action, and the door it will eventually choose.

8 Probe-calibrated belief steering

We fit a binary lakes/rocky probe $p(h) = \sigma(w^\top h + b)$ (held-out accuracy 0.994) and steer the hidden state so that the probe reads a chosen probability, asking whether the *door choice follows the probe probability smoothly*. Two interventions give sharply different answers.

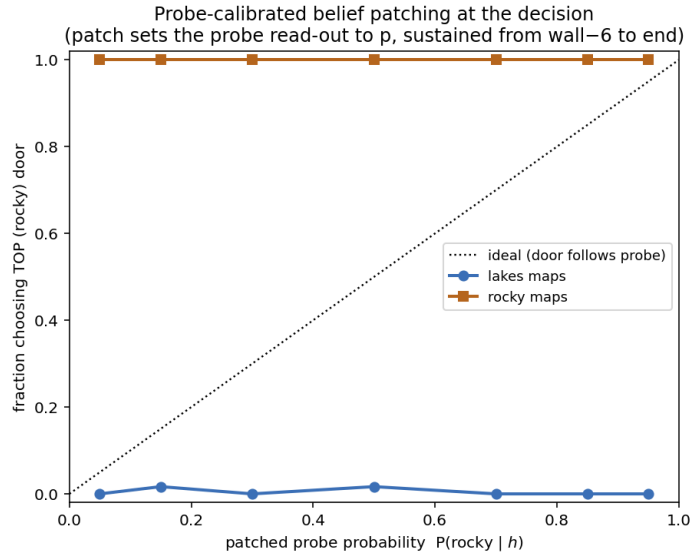


Figure 9: Minimal-norm probe patching is behaviourally inert. From the wall approach (wall-6) to termination the state is patched with $h \leftarrow h + \frac{\text{logit}(p^*) - (w^\top h + b)}{\|w\|^2} w$ — the smallest edit that sets the probe read-out exactly to p^* . Despite the probe reading anything from 0.05 to 0.95, the door choice never moves (lakes $\rightarrow$ bottom, rocky $\rightarrow$ top, at every p^*). Reading out a belief at 0.994 accuracy does not mean the read-out direction is the direction the policy *uses*: a probe-minimal edit misses the causal coordinates entirely (and a late, local patch cannot rewrite a decision whose inputs were integrated over the whole approach).

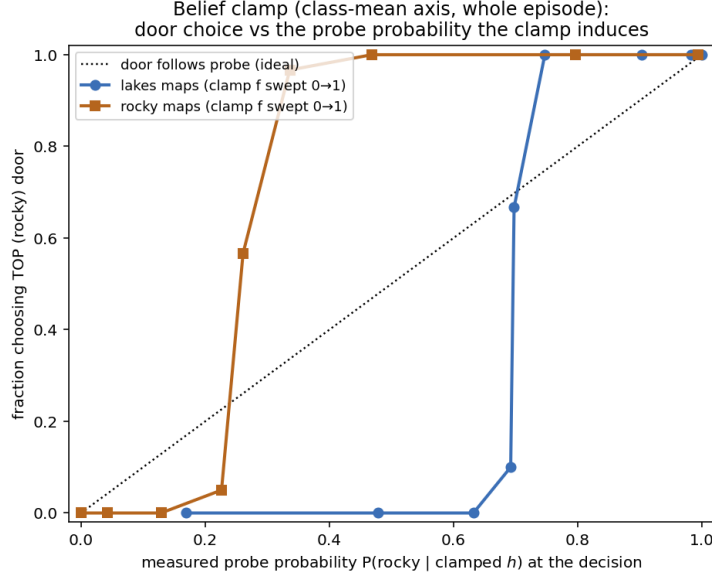


Figure 10: Class-mean belief clamping does move the door — monotonically, but winner-take-all. The working intervention: clamp the state’s projection onto the *time-binned class-mean axis* (lakes↔rocky) at interpolation fractions $f \in [0, 1]$, sustained over the whole episode; for each f we *measure* the probe probability the clamp actually induces at the decision (x-axis) and the resulting door choice (y-axis). Unlike the probe-minimal patch, the door now follows the induced belief across its full range on both map categories — but as a much *steeper* sigmoid than the identity: the policy reads the belief winner-take-all rather than probability-matching, so a probe reading of 0.5 does not produce a 50/50 door split; the behavioural 50/50 point sits at ≈ 0.7 (lakes) and ≈ 0.25 (rocky) in probe units, the probe-space image of the single off-centre decision boundary found in the clamp-fraction psychometric. Two lessons: steering must move the state along the manifold the network traverses (not merely change what a decoder reports), and even then the decoder’s probability scale is not the policy’s.

9 Advance knowledge of the crossing plan

Custom map construction. The planning analysis uses *toy maps* derived from a held-out **balanced** map record: all terrain between the spawn column and the fork wall is flattened to grass, and a single obstacle band is painted at columns 24–26: **water** or **rock**, extending vertically from the spawn row $\pm hh$ with half-height $hh \in \{5, 8, 11\}$ (six families). The wall, passage and doors are kept, and the **balanced** category makes either door correct — so the only decision under study is *how to cross the band*: walk around its edge, or cross through it with the appropriate skill (bridge/tunnel). Each successful episode is labelled by the strategy it actually expressed (skill used vs. no skill), and a logistic probe predicts that label from the hidden state at fixed column distances *before* the obstacle.

Toy maps: flat grass corridor + one obstacle band centred on the spawn row (band = column 24-26, height = spawn $\pm$ hh); wall & doors kept

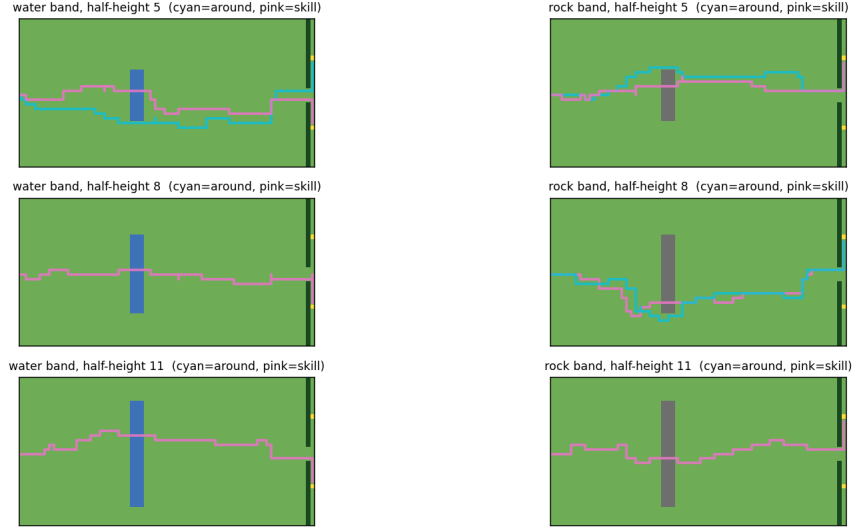


Figure 11: The six toy-map families (water/rock $\times$ half-height 5/8/11) with one example episode of each strategy overlaid (cyan = walks around the band; pink = crosses with bridge/tunnel).

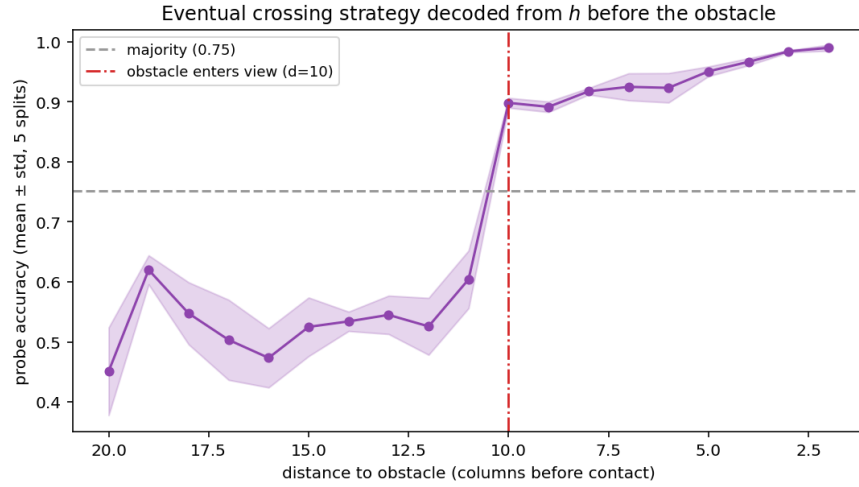


Figure 12: Strategy decodability vs. distance to the obstacle (mean $\pm$ std over 5 episode splits; class-balanced probe). Accuracy sits at chance while the obstacle is outside the agent's 21×21 view, jumps the moment it enters the view radius ($d=10$), and approaches ceiling by contact: the crossing plan is formed on first sight, well before execution.